Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: DESIGN CALCULATION 49 FE FUEL STORAGE FACILITY: LPG BULLET AND MISCELLANEOUS PIPE SUPPORTS DESIGN OF FOUNDATION

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ATTACHMENT 3 MISCELLANEOUS PIPE SUPPORT DESIGN..... 29

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of foundation of LPG Bullet and miscellaneous pipe supports.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to the foundation design of LPG bullet and miscellaneous pipe supports.

4. DESIGN METHODOLOGY & ASSUMPTIONS

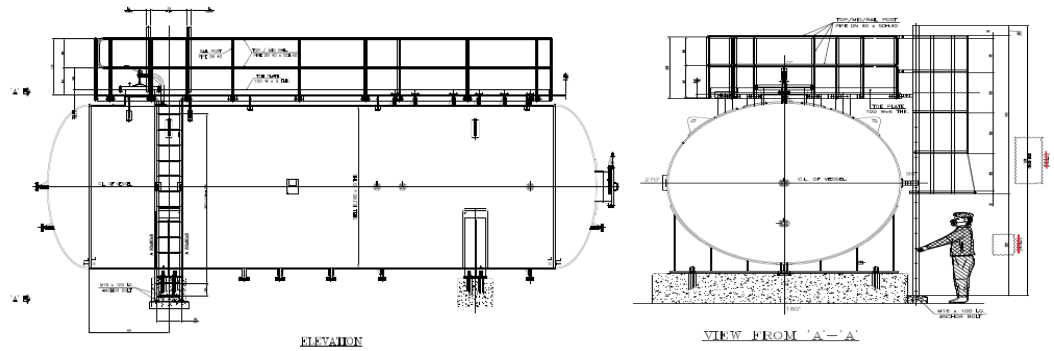
These Buildings LPG bullet GA and saddle layout is taken from available mechanical drawing Foundation loadings are calculated manually. Foundation design is performed using excel sheet with calculated loads.

For miscellaneous pipe supports loadings are calculated based on input provided in overall layout drawing.

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4.1. Codes

[1] SBC-301-2007- Structural Loading & Forces.

[2] SBC-303-2007- Structural Soil & Foundations.

[3] SBC-304-2007- Structural Concrete Structures.

[4] SASO SSA 2 - Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 - Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 - Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 - Building Code Requirements.

[8] PCI –MNL-120-04 - Design Handbook, 6th edition.

[9] SBC-306-2007- Structural – Steel Structures.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

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5. DESIGN LOADS

5.1. Dead Load(DL)

5.1.1. Empty, Operating & Test Load

- Total empty load, De = (1.1 x 275) kN = 302.5 kN
- Total Operating Load. Do = (1.1 x 740) kN = 814 kN
- Total Test load, Dt = (1.1 x 1062) kN = 1170 kN

Equipment loads are obtained from vendor drawing. (See attachment 1).

Note: Tank weight supplied by manufacturer does not include the weight of attached pipe, platform etc. An increase of 10% weight is considered to account for above mentioned items.

5.1.2. Transverse Moment from eccentricity:

Eccentricity considered, e = (basic diameter/2 + 1.5ft) m (based on Process Industry practice guideline)

Hence, e = (3.2/2 + 0.45) m = 2.05 m

- Empty Transverse Moment, MTe = 275 x 0.1 x 2.05 = 56.4 kN.m
- Operating Transverse Moment, MTo = 740 x 0.1 x 2.05 = 151.7 kN.m
- Test transverse Moment, MTt = 1062 x 0.1 x 2.05 = 217.7 kN.m

Note: Based on CG of the equipment shown in drawing (see attachment 1), dead load is distributed equally in fixed and sliding saddle.

5.1.3. Live Loads (LL):

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Access way = **5 kN/m²**

Area of platform attached = 9m x 2m.

Hence, total LL = 5 x 9 x 2 = 90 kN.

Live load distributed equally between fixed and sliding saddle.

5.1.4. Thermal Force (Tf):

(a) Compute sliding force assuming steel to steel friction:

$\mu = 0.4$ (steel to steel coefficient of friction)

Friction on sliding saddle = Ff = 0.4 x 407 = 162.8 kN.

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(b) Compute force required to deflect pier:

Assumed pier size = 3.3m x 0.55m x 1.7m

Moment of inertia, I = (3300 x 550^3)/12 = 4.575 x 10^{10} mm^4

Modulus of elasticity, E = 24.84 kN/m^2

Tank design temperature = 55^0C

Thermal expansion coefficient for carbon steel at 55^0C, ε = 0.0005 mm/mm (see attachment 2).

Hence, thermal growth = εL = 0.0005 x 6000 = 3.05 mm

Thus, T = 3(εL)EI/$2H^3$ = (3 x 3.05 x 24.84 x 4.575 x 10^{10})/(2 x 1700^3) = 1058 kN.

Hence, Ff<T, force on pedestal due to temperature = 162.8 kN.

5.2. Wind Load (WL)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

The Wind Loads calculated here are in accordance with SBC-301-2007-Structural Loading & Forces.

General wind load requirements:

Basic wind speed:	V = 49 m/s (or) 177 kmph (Refer Clause-4.2.6, Doc-MD-216-1000-EG-ST-CRT-1001)
Risk category:	II (As per SBC-301, Table 1.6-1)
Velocity pressure coefficient:	K_d = **0.85** (As per SBC-301, Table 6.4-1)
Exposure category:	D (As per SBC-301, Section 6.4.2.3)
Enclosure classification:	Enclosed buildings
Gust effect factor:	G_f = **0.85** (As per SBC-301, Section-7.2.7.1)
Importance factor:	1 (As per SBC-301, Section-9.5)
Topography factor:	Kzt = 1.0 (As per SBC-301, Section 6.4.3.2)

Thus, q_z = .0473 x 10^{-3} Kz Kzt Kd V^2 I (Ref equation 7.2-12, SBC 301) .

= 1.30 kN/m^2

Wind induced forces = q_z x C_f x G x A_f kN

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5.2.1. Wind Perpendicular to tank:

Tank Diameter (D) = (3160 + 20 x 2) = 3200 mm

Height of tank above ground (h) = 2.9 m

Thus, h/D = 2.9/3.2 = 0.9

D sqrt(q_z) = 3.65

Refer, figure 7.2-15, SBC 301, considering rough surface of tank, Cf = 0.7

Projected area of tank = (11.369 x 3.2) m^2 = 36.381 m^2

Thus, wind force, Wf1 = (1.3 x 0.7 x 0.85 x 36.381) kN

= 28.14 kN (force acting at tank center line)

Thus, Shear on each pier = 28.14/2 kN = 14.7 kN

Moment on pedestal top = (14.7 x 1.8) kN.m = 26.46 kN.m

5.2.2. Wind Parallel to tank:

Projected wind area = (3.14 x 3.2^2/4) m^2 = 8.05 m^2

Cf = 1.2 (Ref fig 7.2-16, SBC 301)

Thus, wind force, Wf2 = (1.3 x 1.2 x 0.85 x 8.05) kN = 10.7 kN

Hence, Shear on each pier = 10.7/2 kN = 5.35 kN

Push/Pull on pedestal top = +/- (10.7 x 1.8/6) kN = +/- 3.21 kN

5.2.3. Wind on Platform:

(a) Wind perpendicular to tank, Wf3 = (1.3 x 1.8 x 9 x 1.1) kN = 23.166 kN

Thus, Shear on each pier = 23.166/2 kN = 11.6 kN

Moment on pedestal top = (11.6 x 1.8) kN.m = 20.88 kN.m

(b) Wind parallel to tank, Wf4 = (1.3 x 1.8 x 1.1 x 2) kN = 5.15 kN

Hence, Shear on each pier = 5.15/2 kN = 2.575 kN

Push/Pull on pedestal top = +/- (2.575 x 1.8/6) kN = +/- 0.8 kN

Platform height = 1.1 m

Platform size = 9m x 2m

Cf = 1.8

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Note: Calculated wind load is less than friction load. Hence, wind load is distributed equally between piers. Table below represents the wind forces on pedestal top:

5.3. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

Hence, seismic base shear Vf = (1.2 x S_{DS} x W)/R = (1.2 x 0.106 x 407/3) kN = 17.3 kN

5.3.1. Seismic Load Perpendicular to tank:

Shear on each pier = 17.3 kN

Moment on pedestal top = (17.3 x 1.8) kN.m = 31.5 kN.m

5.3.2. Seismic Load Parallel to tank:

Hence, Shear on each pier = 17.3kN

Push/Pull on pedestal top = +/- (17.3 x 1.8/6) kN = +/- 5.2 kN

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6. STEEL BEARING PLATE & PEDESTAL SIZING:

Bearing plate width = saddle width +50mm = 450+50 = 500mm

Bearing plate length = Saddle length + 50mm = 3100 + 50 =3150mm

Pier length = bearing plate length + 2x50mm = 3250 mm

Pier width = bearing plate width + 2x50mm = 600 mm

Hence, provide 3300mm x 650mm pedestal size.

7. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +T+LL

D +0.75LL……*not used as non-critical*

D + (W or 0.7E)

D + 0.75W + 0.75LL

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr……*not used as non-critical*

1.2D+ 1.2T + 1.6Lr

1.2D + 1.6W + 1.0Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load, separate load case created for empty, operating and test condition as applicable.

Lr = Live Load

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W = Wind Load

E = Seismic Load

T = Temperature Load

8. DESIGN CALCULATION FOR FOUNDATION

FOUNDATION MKD. F1

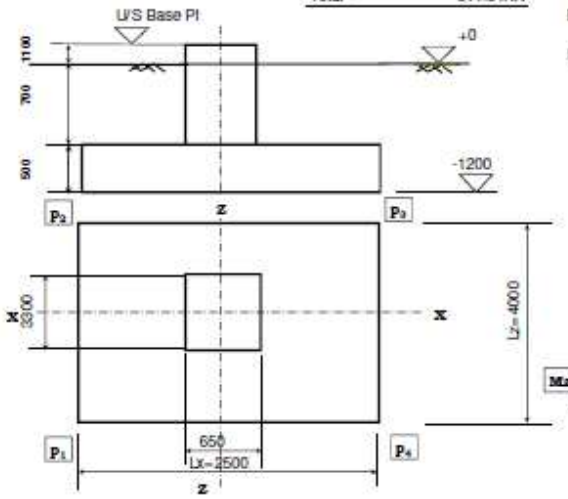
	Along-X	Along-Z
Size of column :	0.65m	3.30m
Size of fdn. raft :	2.50m	4.00m
Thickness of raft :	0.50 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.20 m	
U/S level Base Plate :	1.10 m	
Unit weight of conc :	24 kN/m³	
Unit weight of soil :	18 kN/m³	

Calculation of Weight of Foundation :

Weight of raft & Col.= (2.5*4*0.5*24)+(0.65*3.3*1.8*24) = 212.66KN

Weight of soil = ((2.5*4)-(0.65*3.3))*0.7*18 = 98.97KN

Total = 311.64KN



Design Input:

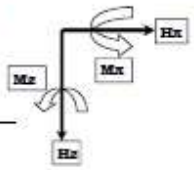
Characteristic strength of concrete, f'c =	35 MPa
Yield stress of steel reinforcement, fy =	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provided along Z direction at bottom =	16mm
Minimum percentage of reinforcement at bottom =	0.18%

Effective depth of slab along X direction = 442mm = 0.442m

Effective depth of slab along Z direction = 426mm = 0.426m

Clear projection along Z direction = 0.35m

Clear projection along X direction = 0.925m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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FOUNDATION MKD. F1

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Empty Load (De)	152.00	0.00	0.0	0.0	28.7
2	Operating Load (Do)	407.00	0.00	0.0	0.0	75.9
3	Test Load (Dt)	585.00	0.00	0.0	0.0	109.0
4	Temparature Load (T)	0.00	163.00	0.0	0.0	0.0
5	Wind Long (WL)	4.50	8.00	0.0	0.0	0.0
6	Wind Transverse (WT)	0.00	0.00	0.0	17.3	47.4
7	Seismic Long (EL)	5.22	17.40	0.0	0.0	0.0
8	Seismic Transverse (ET)	0.00	0.00	0.0	17.4	31.3
9	Live Load (LL)	45.00	0.00	0.0	0.0	0.0

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
101	Do+T	407.0	163.0	0.0	0.0	75.9	718.64	163.00	374.90	0.00	75.85
102	Dt+T	585.0	163.0	0.0	0.0	109.0	896.64	163.00	374.90	0.00	109.00
103	Do+WL	411.5	8.0	0.0	0.0	75.9	723.14	8.00	18.40	0.00	75.85
104	Dt+WL	589.5	8.0	0.0	0.0	109.0	901.14	8.00	18.40	0.00	109.00
105	Do+WT	407.0	0.0	0.0	17.3	123.2	718.64	0.00	0.00	17.30	162.99
106	Dt+WT	585.0	0.0	0.0	17.3	156.4	896.64	0.00	0.00	17.30	196.14
107	Do+0.7EL	410.7	12.2	0.0	0.0	75.9	722.29	12.18	28.01	0.00	75.85
108	Dt+0.7EL	588.7	12.2	0.0	0.0	109.0	900.29	12.18	28.01	0.00	109.00
109	Do+0.7ET	407.0	0.0	0.0	12.2	97.8	718.64	0.00	0.00	12.18	125.79
110	Dt+0.7ET	585.0	0.0	0.0	12.2	130.9	896.64	0.00	0.00	12.18	158.94
111	0.6De+WL	86.7	8.0	0.0	0.0	17.2	398.34	8.00	18.40	0.00	17.22
112	0.6De+WT	91.2	0.0	0.0	17.3	64.6	402.84	0.00	0.00	17.30	104.36
113	0.6De+EL	86.0	17.4	0.0	0.0	17.2	397.62	17.40	40.02	0.00	17.22
114	0.6De+ET	91.2	0.0	0.0	17.4	48.5	402.84	0.00	0.00	17.40	88.56
115	Do+LL+T	452.0	163.0	0.0	0.0	75.9	763.64	163.00	374.90	0.00	75.85
116	Do+0.75WL+0.75LL	444.1	6.0	0.0	0.0	75.9	755.77	6.00	13.80	0.00	75.85
117	Do+0.75WT+0.75LL	440.8	0.0	0.0	13.0	111.4	752.39	0.00	0.00	12.98	141.21

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201	1.4(Do+T)	881.44	524.86	106.19	88.14	125.97	15.93	250.00	250.00	0.00	0.00	228.40	250.00	0.00	125.00	125.00
202	1.4(Dt+T)	1130.64	524.86	152.60	113.06	125.97	22.89	270.00	270.00	9.99	0.00	248.40	270.00	5.00	140.00	135.00
203	1.2Do+1.6WL	807.24	29.44	91.02	80.72	7.07	13.65	101.44	74.14	87.31	60.01	79.84	87.79	73.66	94.38	67.08
204	1.2Dt+1.6WL	1020.84	29.44	130.80	102.08	7.07	19.62	128.77	89.53	114.64	75.40	107.17	109.15	95.02	121.71	82.47
205	1.2Do+1.6WT	800.04	0.00	230.44	80.00	0.00	34.57	114.57	45.44	114.57	45.44	92.97	80.01	80.01	114.57	45.44
206	1.2Dt+1.6WT	1013.64	0.00	270.22	101.36	0.00	40.53	141.90	60.83	141.90	60.83	120.30	101.37	101.37	141.90	60.83
207	1.2Do+EL	805.26	40.02	91.02	80.53	9.60	13.65	103.78	76.48	84.57	57.27	82.18	90.13	70.92	94.18	66.88
208	1.2Dt+EL	1018.86	40.02	130.80	101.89	9.60	19.62	131.11	91.87	111.90	72.66	109.51	111.49	92.28	121.51	82.27
209	1.2Do+ET	800.04	0.00	162.36	80.00	0.00	24.35	104.36	55.65	104.36	55.65	82.76	80.01	80.01	104.36	55.65
210	1.2Dt+ET	1013.64	0.00	202.14	101.36	0.00	30.32	131.69	71.04	131.69	71.04	110.09	101.37	101.37	131.69	71.04
211	0.9De+1.6WL	441.24	29.44	25.83	44.12	7.07	3.87	55.06	47.32	40.93	33.18	33.46	51.19	37.06	48.00	40.25
212	0.9De+1.6WT	448.44	0.00	97.17	44.84	0.00	14.58	59.42	30.27	59.42	30.27	37.82	44.85	44.85	59.42	30.27
213	0.9De+EL	443.22	40.02	25.83	44.32	9.60	3.87	57.80	50.05	38.59	30.84	36.20	53.93	34.72	48.20	40.45
214	0.9De+ET	448.44	0.00	97.17	44.84	0.00	14.58	59.42	30.27	59.42	30.27	37.82	44.85	44.85	59.42	30.27
215	1.2Do+1.2T+1.6LL	872.04	449.88	91.02	87.20	107.97	13.65	250.00	200.00	0.00	0.00	228.40	225.00	0.00	125.00	100.00
216	1.2Do+1.6WL+0LL	852.24	29.44	91.02	85.22	7.07	13.65	105.94	78.64	91.81	64.51	84.34	92.29	78.16	96.88	71.58
217	1.2Do+1.6WT+1.0LL	845.04	0.00	230.44	84.50	0.00	34.57	119.07	49.94	119.07	49.94	97.47	84.51	84.51	119.07	49.94
	Max :	1131	525	270												
	Min :	398	0	17												

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 178.40 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 198.40 kN/m²
Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 248.40 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 268.40 kN/m²

Sample calculation of values of base pressure:

Load case : Dt+T

Maximum pressure, p_1 (pmax) = 89.66+89.98+16.35= 195.99 kN/m²
p_2= 89.66+89.98-16.35= 163.29 kN/m²
p_3= 89.66-(89.98+16.35)= -16.66 kN/m²
p_4= 89.66-89.98+16.35= 16.04 kN/m²
Maxm pressure excluding weight of soil (p_{max}-γh)= 200.00-1.2*18= 178.40 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (200.00+200.00)/2= 179.64 kN/m²
Minimum average Pressure along X, $p_{avx(minx)}$ = (0.00+16.04)/2= -0.31 kN/m²
Maximum average Pressure along Y, $p_{avy(max)}$ = (200.00+16.04)/2= 106.01 kN/m²
Minimum average Pressure along Y, $p_{avy(minx)}$ = (0.00+200.00)/2= 73.31 kN/m²

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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.65+2*0.5*0.426)+(3.3+2*0.5*0.426))*2

= 9.604 m

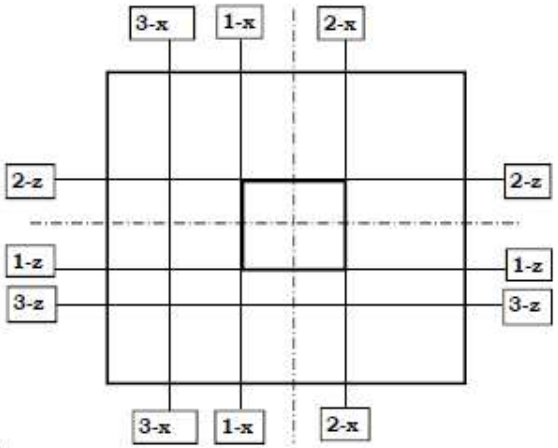
Punching Area= ((0.65+2*0.5*0.426)*(3.3+2*0.5*0.426))

= 4.01 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*24+0.7*18)

= 24.60 kN/m^2

Area of footing, A_f= 10.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Condition								
201	1.4(Do+T)	157.50	92.50	201.70	83.24	2.67	97.20	380.63
202	1.4(Dt+T)	171.95	103.05	218.80	91.00	5.60	106.16	529.90
203	1.2Do+1.6WL	82.56	78.89	85.06	26.29	21.73	29.86	336.18
204	1.2Dt+1.6WL	103.92	100.25	106.42	35.43	30.87	40.18	464.13
205	1.2Do+1.6WT	80.01	80.01	80.01	23.71	23.71	26.76	331.87
206	1.2Dt+1.6WT	101.37	101.37	101.37	32.84	32.84	37.08	459.82
207	1.2Do+EL	83.02	78.03	86.42	27.02	20.83	30.75	335.00
208	1.2Dt+EL	104.38	99.39	107.78	36.16	29.97	41.07	462.94
209	1.2Do+ET	80.01	80.01	80.01	23.71	23.71	26.76	331.87
210	1.2Dt+ET	101.37	101.37	101.37	32.84	32.84	37.08	459.82
211	0.9De+1.6WL	45.96	42.29	48.46	10.63	6.08	12.18	116.95
212	0.9De+1.6WT	44.85	44.85	44.85	8.66	8.66	9.78	121.26
213	0.9De+EL	46.82	41.83	50.22	11.53	5.34	13.27	118.13
214	0.9De+ET	44.85	44.85	44.85	8.66	8.66	9.78	121.26
215	1.2Do+1.2T+1.6LL	141.75	83.25	181.53	73.86	1.35	86.30	375.00
216	1.2Do+1.6WL+0LL	87.06	83.39	89.56	28.21	23.66	32.03	363.14
217	1.2Do+1.6WT+1.0LL	84.51	84.51	84.51	25.63	25.63	28.94	358.82

Sample calculation of BM & SF

Load case : 201

p_{1x} =	250-(250-0)*0.925/2.5 =	157.50 kN/m²	Overhang =	0.925 m
p_{2x} =	0+(250-0)*0.925/2.5 =	92.50 kN/m²	Effective depth =	0.442 m
p_{3x} =	250-(250-0)*(0.925-0.442)/2.5 =	201.70 kN/m²		

BM_{1x} =	(157.50-24.6)*0.925*0.925/2+0.5*(250-157.50)*0.925*0.925*0.67 =	83.24 kN-m/m
BM_{2x} =	(92.50-24.6)*0.925*0.925/2-0.5*(-0+92.50)*0.925*0.925*0.67 =	2.67 kN-m/m
SF_{3x} =	(201.70-24.6)*(0.925-0.442)+0.5*(250-201.70)*(0.925-0.442) =	97.20 kN/m
$S.F_{(punching)}$ =	(10-4.01)*(88.14-24.6) =	380.63 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	91.00	0.47	0.62	0.001	660.58	795.60	16ϕT	252.72	300	200	1005.31
At top	0.00	0.00	0.00	0.000	0.00	795.60	16ϕT	252.72	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 106.16 kN/m

Shear stress , tv= (106.16*1000)/(1000*442)

= 0.24 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu=	529.90 kN	
Strength of Concrete in shear Vc1 =	46685.48 kN	Table 22.6.5.2_ (a)
Strength of Concrete in shear Vc2 =	6111.91 kN	Table 22.6.5.2_ (b)
Strength of Concrete in shear Vc3 =	8371.19 kN	Table 22.6.5.2_ (c)
Design Vc=	6111.91 kN	
Allowable shear stress, $(\phi\ Vc)/bd$ =	1.08 N/mm2	

Design punching shear stress, τ = (529.9*1000)/(9604*442)

= 0.12 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m^2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m^2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
201	1.4(Do+T)	125.00	125.00	125.00	6.15	6.15	-7.63
202	1.4(Dt+T)	139.56	135.44	140.10	7.06	6.77	-8.77
203	1.2Do+1.6WL	91.99	69.47	94.90	4.23	2.65	-5.32
204	1.2Dt+1.6WL	118.28	85.90	122.46	5.88	3.61	-7.41
205	1.2Do+1.6WT	108.52	51.49	115.88	5.39	1.40	-6.89
206	1.2Dt+1.6WT	134.81	67.92	143.44	7.04	2.36	-8.97
207	1.2Do+EL	91.79	69.27	94.70	4.21	2.64	-5.31
208	1.2Dt+EL	118.08	85.70	122.26	5.87	3.60	-7.39
209	1.2Do+ET	100.10	59.91	105.29	4.80	1.99	-6.10
210	1.2Dt+ET	126.38	76.35	132.84	6.45	2.95	-8.18
211	0.9De+1.6WL	47.32	40.93	48.15	1.42	0.97	-1.78
212	0.9De+1.6WT	56.87	32.82	59.97	2.08	0.40	-2.67
213	0.9De+EL	47.52	41.13	48.35	1.43	0.98	-1.80
214	0.9De+ET	56.87	32.82	59.97	2.08	0.40	-2.67
215	1.2Do+1.2T+1.6LL	122.81	102.19	125.48	6.11	4.66	-7.65
216	1.2Do+1.6WL+0LL	96.49	73.97	99.40	4.50	2.93	-5.66
217	1.2Do+1.6WT+1.0LL	113.02	55.99	120.38	5.66	1.67	-7.23

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Sample calculation of BM & SF

Load case : 201

p_{1z} = 125-(125-125)*0.35/4= 125.00 kN/m^2

p_{2z} = 125+(125-125)*0.35/4= 125.00 kN/m^2

p_{3z} = 125-(125-125)*(0.35-0.426)/4= 125.00 kN/m^2

Overhang = 0.35 m

Effective depth = 0.426 m

BM_{1z} = (125.00-24.6)*0.35*0.35/2+0.5*(125-125.00)*0.35*0.35*0.67 = 6.15 kN-m/m

BM_{2z} = (125.00-24.6)*0.35*0.35/2-0.5*(-125+125.00)*0.35*0.35*0.67 = 6.15 kN-m/m

SF_{3z} = (125.00-24.6)*(0.35-0.426)+0.5*(125-125.00)*(0.35-0.426) = -7.63 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	7.06	0.039	0.052	0.0001	54.63	766.80	16φT	262.21	300	200	1005.3
At top	0.00	0.000	0.000	0.0000	0.00	766.80	16φT	262.21	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm2

Shear at a distance 'd' from face of column, V_u= -5.32 kN/m

Shear stress , τ_v= (-5.32*1000)/(1000*426)

= -0.01 N/mm2 O.K.

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16φT	200	c/c
	At Top	16φT	200	c/c
Along Z Direction :				
	At Bottom	16φT	200	c/c
	At Top	16φT	200	c/c

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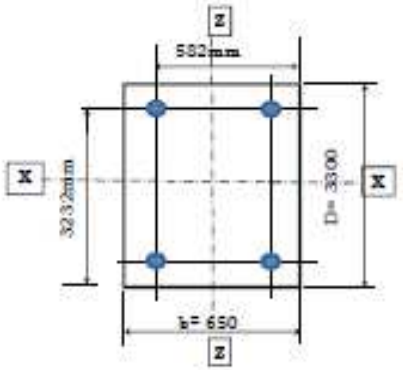
9. DESIGN CALCULATION FOR COLUMN

COLUMN MKD C1

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	16mm
Dia. of link=	10mm
Effective cover =	60mm
Grade of Concrete (f_c) =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.15%
Max. Steel Ratio	4.00%
Self weight of Column =	92.66kN

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) =	1800mm
Unsupported length of column about Z (l_{uz}) =	1800mm

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio,	$K l_{ux} / r_x$ =	3.64	< 22 SECTION 6.2.5a	Radius of gyration r_x=	990	mm
	$K l_{uz} / r_z$ =	18.46	< 22 SECTION 6.2.5a	Radius of gyration r_z=	195	mm

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Reinforcement: Provide,

4-Nos	16 mm Dia
26-Nos	16 mm Dia
2-Nos	16 mm Dia

804.57 Sq mm (At Corner)
5229.71 Sq mm (Along Z axis)
402.29 Sq mm (Along Y axis)

A_{sc} pvd. = 6436.57 Sq mm

Total A_{sc} = 0.30 % ρ_g = 0.003

$\beta = P_n / P_o$ (range is 0.55 to 0.70)

Where, $P_o = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 6632.562 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
1.4(Do+T)	0.050	0.050	2439.9	12387.4	0.55	1.16	0.21	O.K
1.4(Dt+T)	0.050	1.050	2439.9	260134.9	0.55	1.16	0.21	O.K
1.2Do+1.6WL	0.050	2.050	2439.9	507882.4	0.55	1.16	0.01	O.K
1.2Dt+1.6WL	0.050	3.050	2439.9	755629.9	0.55	1.16	0.01	O.K
1.2Do+1.6WT	0.050	4.050	2439.9	1003377.4	0.55	1.16	0.00	O.K
1.2Dt+1.6WT	0.050	5.050	2439.9	1251124.9	0.55	1.16	0.00	O.K
1.2Do+EL	0.050	6.050	2439.9	1498872.4	0.55	1.16	0.01	O.K
1.2Dt+EL	0.050	7.050	2439.9	1746619.9	0.55	1.16	0.01	O.K
1.2Do+ET	0.050	8.050	2439.9	1994367.4	0.55	1.16	0.00	O.K
1.2Dt+ET	0.050	9.050	2439.9	2242114.9	0.55	1.16	0.00	O.K
0.9De+1.6WL	0.050	10.050	2439.9	2489862.4	0.55	1.16	0.01	O.K
0.9De+1.6WT	0.050	11.050	2439.9	2737609.9	0.55	1.16	0.00	O.K
0.9De+EL	0.050	12.050	2439.9	2985357.4	0.55	1.16	0.01	O.K
0.9De+ET	0.050	13.050	2439.9	3233104.9	0.55	1.16	0.00	O.K
1.2Do+1.2T+1.6LL	0.050	14.050	2439.9	3480852.4	0.55	1.16	0.17	O.K
1.2Do+1.6WL+0LL	0.050	15.050	2439.9	3728599.9	0.55	1.16	0.01	O.K
1.2Do+1.6WT+1.0LL	0.050	16.050	2439.9	3976347.4	0.55	1.16	0.00	O.K

Links, 10 mm dia @ 250 C/C

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REFERENCES:

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Document. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1.

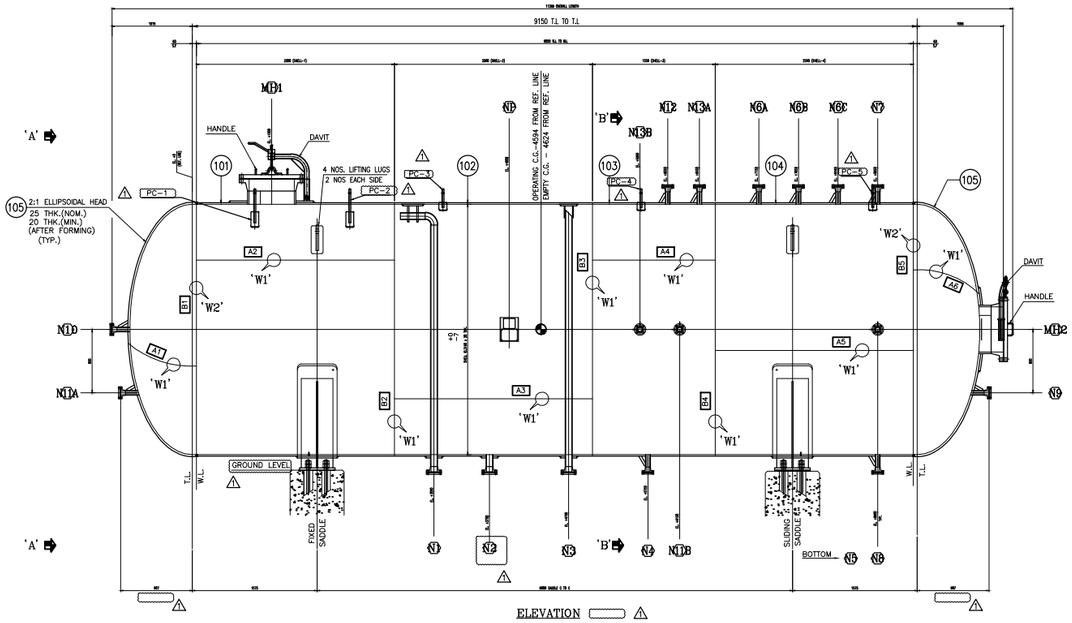
ATTACHMENT 1 LPG BULLET DRAWING

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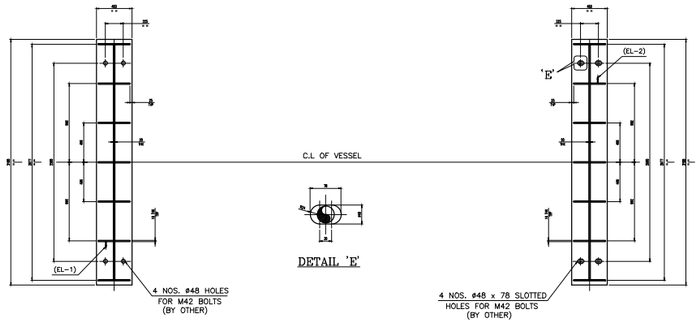
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ELEVATION ⚠

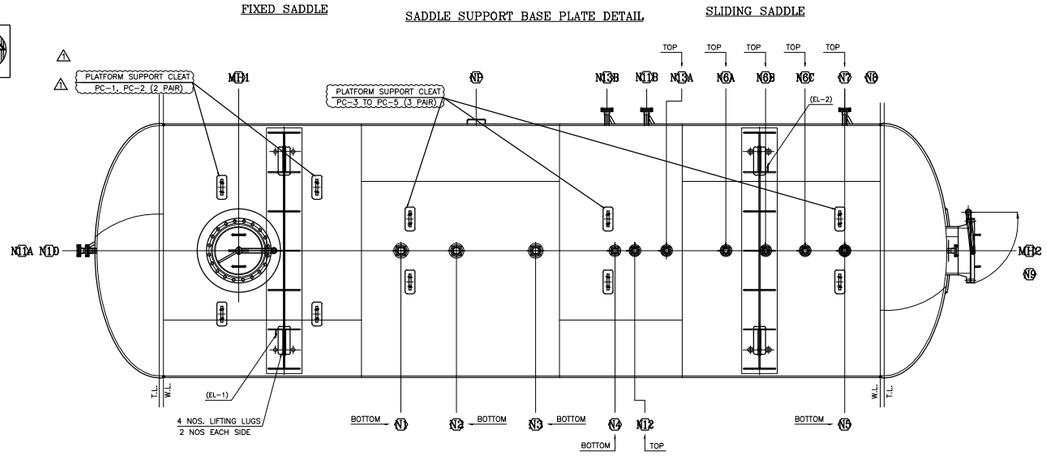


DETAIL 'E'

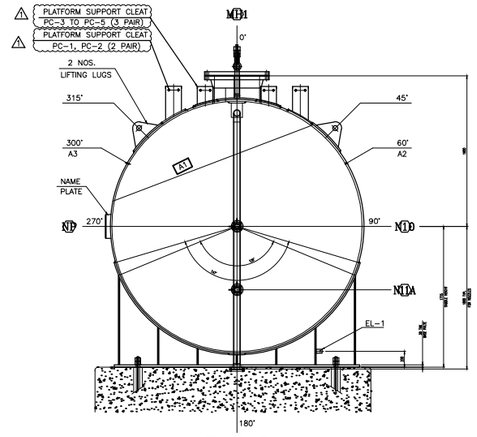
FIXED SADDLE

SADDLE SUPPORT BASE PLATE DETAIL

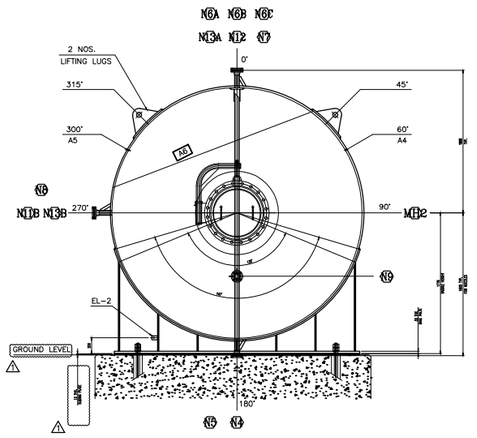
SLIDING SADDLE



PLAN ⚠

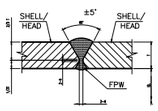


VIEW FROM 'A'–'A'



VIEW FROM 'B'–'B'

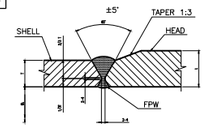
'W1'



A1 TO A6 &
B2 TO B4
(TYPE NO.1)

TYP. WELD DETAIL FOR
SHELL TO SHELL &
HEAD TO HEAD JOINT

'W2'



B1 & B5
(TYPE NO.1)

SHELL TO HEAD WELD DETAIL

DESIGN DATA

			CLAUSE
CONSTRUCTION CODE		ASME SECT. VIII DIV. 1 EDITION 2019	
CODE CASES		NIL	
CERTIFICATION MARK		YES, ASME U-DESIGNATOR	
NATIONAL BOARD REGISTRATION		NOT REQUIRED	
MANUFACTURER SERIAL No.		PVI-199	
INSPECTION		PVI / AI / CLIENT	
EQUIPMENT No.		–	
QUANTITY OF EQUIPMENT	Nos.	01	
PRESSURE: DESIGN (INT/EXT)	MPa(g) (bar(g))	1.45 (14.5) / N.A.	UG-21
PRESSURE: OPERATING (MAX/MIN)	MPa(g) (bar(g))	– / –	
PRESSURE: HYDROTEST AT TOP OF VESSEL SHOP & SITE	MPa(g) (bar(g)) [Kg/Cm2(g)]	2.01 (20.1) [20.5 Kg/Cm2] SHOULD NOT ⚠ EXCEED 21.5 Kg/Cm2	(USER DEFINED) WHICH IS GREATER THAN UG-99(b) AND NOTE-35
PRESSURE: MAWP (INTERNAL) (HOT & CORRODED)	MPa(g) (bar(g))	1.45 (14.5) @ 55°C (SAME AS DESIGN PRESSURE)	UG-98
PRESSURE: MAWP (EXTERNAL)	MPa(g) (bar(g))	NOT APPLICABLE	
TEMPERATURE: DESIGN (INT/EXT)	°C	55 / NA	UG-20
TEMPERATURE: OPERATING (MAX/MIN)	°C	– / –	
TEMPERATURE: MDMT	°C	-29°C @ 1.45 MPa(g) [14.5 bar(g)]	UG-20
METAL TEMPERATURE DURING HYDROTEST	°C	NOT LESS THAN 17 & NOT BE MORE THAN 48	UG-99(h)
OPERATING MEDIUM		LPG	
CAPACITY	Cu. m	80.04	
MEDIUM DENSITY	Kg/Cu. m	580	
CORROSION ALLOWANCE (INT/EXT)	(mm)	1.5/NIL	
RADIOGRAPHY MARKING		[RT-1] (AS PER UG-116(e)(1),UW-11(a)	REFER TABLE-1
POST WELD HEAT TREATMENT (PWHT)		YES, FOR COMPLETE VESSEL (AS PER CLIENT REQUIREMENT) UW-2 & UCS-56	REFER NOTE-4.2
POST FORMING HEAT TREATMENT (PFHT)		NO (EXEMPTED AS PER UG-79 & UCS-79)	
IMPACT TESTING		NO (EXEMPTED AS PER UG-20(f), UCS-66 (a),UCS-66(c) FIG UCS-66 GENERAL Note(c) & UCS-67 (a)	
SPECIAL SERVICE		NO (AS PER UW-2)	
INSPECTION OPENING REQUIREMENT		YES (PROVIDED AS PER UG-46(f)(3), UG-46(g)(1))	
WIND LOAD		⚠ ASCE-2016, WIND SPEED -177 km/hr OPEN TERRAIN, SOIL TYPE-MODERATELY SMOOTH.	UG-22
SEISMIC LOAD		⚠ NA	
INSTALLATION		HORIZONTAL	
INSULATION	TYPE / THK	NA	
FIREPROOFING	TYPE / THK	NA	
TEST MEDIA		WATER	
HYDRO TEST POSITION (SHOP/SITE)		HORIZONTAL/HORIZONTAL	
DRAIN OPENING REQUIREMENT		PROVIDED AS PER UG-25(f)	
PRESSURE SAFETY VALVE / INSTRUMENTS		TO BE PROVIDED BY UNIGAZ	
LOCATION OF INSTALLATION		⚠ MMGP PLANT	

WEIGHTS IN Kgs (FOR ONE EQUIPMENT ONLY)

FABRICATED CONDITION	EMPTY CONDITION	OPERATING CONDITION	HYDROTEST CONDITION
⚠ (~24830 Kg.	~27500 Kg.	~73900 Kg.	~106200 Kg.)

BILL OF MATERIAL (FOR ONE EQUIPMENT ONLY)

PART NO.	DESCRIPTION	QTY.	MATERIAL	SIZE	REMARKS
105	2:1 ELLIPSOIDAL HEAD	02	SA-516 GR. 70N	BLANK DIA 3797 x 25(NOM)/20(MIN.)	–
104	SHELL-4	01	SA-516 GR. 70N	CIRC. 9990 x 2500 x 20 THK.	–
103	SHELL-3	01	SA-516 GR. 70N	CIRC. 9990 x 1550 x 20 THK.	–
102	SHELL-2	01	SA-516 GR. 70N	CIRC. 9990 x 2500 x 20 THK.	–
101	SHELL-1	01	SA-516 GR. 70N	CIRC. 9990 x 2500 x 20 THK.	–

LEGENDS :–

- (xx) DENOTES PART No.
- (Nx) DENOTES NOZZLE No.
- (xx) DENOTES WELD No.
- ⚠ DENOTES REVISION No.

T.L. – TANGENT LINE
W.L. – WELD LINE
N.A. – NOT APPLICABLE
NP – NAME PLATE
EL-X – EARTHING LUG NUMBER
SF – STRAIGHT FACE
EL – ELEVATION
ID. – INSIDE DIAMETER
C.G. – CENTER OF GRAVITY
FPW – FULL PENETRATION WELD
HDG – HOT DIP GALVANIZED.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	---	MNP	---	QRS	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER: 2006-UNI-DWG-076 | REVISION: 01

	DATE		DATE		DATE	
DRAWN	GB	CHECKED	NM	APPROVED	SKR	
	26.09.2020		26.09.2020		26.09.2020	

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_V_19163_001_R1 | REVISION: 01

OUTOTEC DRAWING NUMBER: 900707__GAU01__AG50__MTL01__49003 | DOCUMENT ID: OU100688548 | REVISION: 00

DRAWN	---	26.09.2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	MNP	26.09.2020	GENERAL ARRANGEMENT FUEL OIL STORAGE
ENGINEER	---	26.09.2020	FUEL OIL STORAGE TANK
APPROVED	QRS	26.09.2020	LPG BULLET
EMPLOYER			
SCALE	1:XX	A0	DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1003 / REV: A01

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GENERAL NOTES

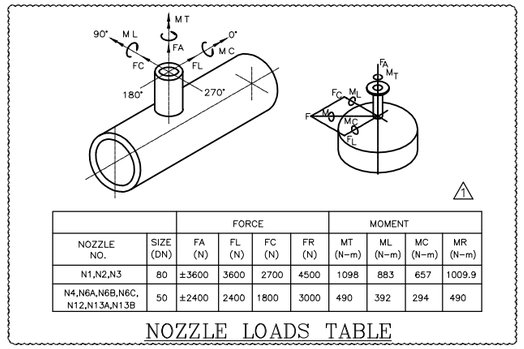
1.	**GENERAL :**
1.1	ALL DIMENSIONS ARE IN MILLIMETER UNLESS OTHERWISE STATED.
1.2	ALL FLANGE BOLT HOLES SHALL STRADDLE NORMAL CENTRE LINE OF VESSEL UNLESS OTHERWISE STATED.
1.3	UNLESS OTHERWISE INDICATED, NOZZLE PROJECTIONS ARE MEASURED FROM VESSEL CENTERLINE & T.L.
1.4	ALL NOZZLES OF LESS THAN DN 50 SHALL BE STIFFENED WITH TWO GUSSET PLATES PLACED AT 90° APART.
1.5	ALL MENTIONED FILLET SIZES ARE AT LEG SIDE.
2.	**WELDING :**
2.1	WELDING OF ALL PRESSURE PARTS SHALL BE FULL PENETRATION, THIS SHALL BE DONE BY BACK CHIPPING TO SOUND METAL AND REWELDING ON THE OTHER SIDE. WHEN OTHER SIDE IS NOT ACCESSIBLE ATLEAST THE ROOT RUN AND HOT RUN SHALL BE DONE BY GTAW WELDING TO ENSURE COMPLETE PENETRATION AND SUBSEQUENT RUNS SHALL BE DONE BY OTHER WELDING PROCESS.
2.2	NO WELDING IS PERMITTED AFTER HYDROTEST.
2.3	THE DISSIMILAR THICKNESS TO BE JOINED BY SMOOTH TRANSITION OF 1:3 TAPER (MINIMUM) AS PER CODE PARA UW-9 & UW-33
2.4	UNLESS OTHERWISE SPECIFIED, THE FILLET WELD THROAT SIZE FOR NON PRESSURE PARTS TO BE 0.7 TIMES OF LEAST THK. OF PARTS TO BE JOINED.
3.	**FABRICATION :**
3.1	ALL SHARP CORNERS TO BE ROUNDED OFF TO A RADIUS OF 3MM.
3.2	ALL NOZZLE INSIDE EDGES EXCEPT NOZZLES WITH INSIDE PROJECTION SHALL BE FLUSHED WITH INTERNAL SURFACE OF VESSEL AND ROUNDED OFF WITH 3MM RADIUS.
3.3	ALL REINFORCEMENT PADS AT NOZZLES, SUPPORTS ETC. SHALL BE PROVIDED WITH A VENT HOLE AT THE TOP-MOST LOCATION, WHICH SHALL BE TESTED WITH AIR AT 1.0 [kgf/cm^2(g)] AND CHECKED WITH SOAP SOLUTION FOR LEAKAGES IF ANY. SUCH VENT HOLES SHALL BE PLUGGED WITH EITHER HARD GREASE OR CAP AFTER SATISFACTORY TESTING.
4.	**INSPECTION & NDE :**
4.1	VISUAL EXAMINATION (INTERNAL AND EXTERNAL) AFTER COMPLETION OF WELDING TO CHECK COMPLIANCE TO UW-35 SHALL BE PERFORMED FOR BOTH LONG'L SEAM (CAT. A) AND CIR'C SEAM (CAT. B) WELDS & AS PER UW-16 FOR D WELDS. IF INSPECTION FROM INSIDE IS NOT ACCESSIBLE THEN REMOTE VISUAL INSPECTION CAN BE CARRIED OUT.
4.2	PWHT FOR COMPLETE VESSEL (STRESS RELIEVING) CAN BE CARRIED OUT AS PER FOLLOWING CYCLE. SOAKING TEMPERATURE 620°C ± 10°C SOAKING TIME 60 MINUTES (MINIMUM) HEATING RATE 70 DEG C PER HOUR FROM 300° C COOLING RATE 60 DEG C PER HOUR UPTO 300°C
5.	**MATERIAL :**
5.1	MATERIAL FOR ALL PERMANENT WELDED ATTACHMENTS TO PRESSURE PARTS SHALL BE OF THE SAME GRADE AS THE SHELL MATERIAL.
5.2	ALL C.S. MATERIALS SHALL BE NORMALIZED CONDITION.
5.3	FORGING SHALL BE SUPPLIED IN NORMALIZED CONDITION.
6.	**TOLERANCE :**
6.1	CODE REQUIRED TOLERANCES ON VESSEL SHALL BE AS PER CODE PARA UG-80(SHELL) & UG-81 (HEAD) FOR OUT OF ROUNDNESS, UW-33(ALIGNMENT) & UW-35(REINFORCEMENT) WHILE FABRICATION AND WELDING FOR OTHER VESSEL FABRICATION TOLERANCE SHALL BE AS PER STD. DRAWING.
7.	**TRANSPORTATION :**
7.1	BEFORE SHIPPING THE VESSEL MUST BE INTERNALLY CLEANED, DRAINED & DRIED.
7.2	ALL NOZZLES NOT SUPPLIED WITH BLIND FLANGES SHALL BE PROTECTED BY WOODEN BLANKS/ DUMMY METAL COVER, TEMPORARY FASTENERS AND GASKETS FOR TRANSPORTATION PURPOSE
7.3	EQUIPMENT SHALL BE WRAPPED WITH PLASTIC STRETCH FILM FOLLOWED BY HDPE CLOTH STICHED AROUND.
7.4	LOOSE SMALL ITEMS SHALL BE SUPPLIED IN WOODEN BOX.
8.	**TESTING :**
8.1	REMOVE ALL WELD SPLATTER, BURRS, SLAG AND SCALE FROM INTERIOR AND EXTERIOR OF VESSEL PRIOR OF HYDROTEST.
8.2	GASKET USED FOR HYDROSTATIC TEST SHALL BE SAME AS THAT OF SERVICE GASKET.
8.3	ALLOWABLE VARIATION IN HYDROTEST PRESSURE IS [1 kg/cm^2(g)] POSITIVE AND 0 NEGATIVE.
8.4	WATER USED FOR HYDROTESTING SHALL HAVE CHLORIDE CONTENT LESS THAN 25 ppm.
9.	**PAINTING :**
9.1	SURFACE PREPARATION – ALL EXTERNAL UNMACHINED SURFACE SHALL BE SHOT BLASTED TO 2 1/2.
9.2	PAINTING SHALL BE AS PER APPROVED PAINTING PROCEDURE (FINISH SHADE RAL 9010).

APPLICABLE LOADING AS PER UG-22

SR.NO.	DESCRIPTION	APPLICABLE
(a)	(1) INTERNAL PRESSURE	YES
	(2) EXTERNAL PRESSURE	NO
(b)	WEIGHT OF VESSEL, NORMAL CONTENTS UNDER OPERATING AND TEST CONDITIONS	YES
(c)	SUPERIMPOSED STATIC REACTIONS FROM WEIGHT OF ATTACHED EQUIPMENTS SUCH AS MOTORS, MACHINERY, OTHERS, VESSELS, LININGS AND INSULATION	YES
(d-1)	THE ATTACHMENT OF INTERNALS.	NO
(d-2)	THE ATTACHMENT OF VESSEL SUPPORTS	YES
(e)	CYCLIC AND DYNAMIC REACTIONS DUE TO PRESSURE OR THERMAL VARIATIONS, OR FROM EQUIPMENT MOUNTED ON A VESSEL AND MECHANICAL LOADINGS.	NO
(f)	WIND LOADING	YES
	SEISMIC LOADING	NO
	SNOW LOADING	NO
(g)	IMPACT REACTIONS SUCH AS THOSE DUE TO FLUID SHOCK	NO
(h)	TEMPERATURE GRADIENT AND DIFFERENTIAL THERMAL EXPANSION	NO
(i)	ABNORMAL PRESSURES, SUCH AS THOSE CAUSED BY DEFLAGRATION	NO
(j)	TEST PRESSURE AND COINCIDENT STATIC HEAD ACTING DURING TEST (UG-99)	YES

NOZZLE MARK	SERVICE	DN SIZE	SCH/THK.	RATING / AS PER ASME B16.5	TYPE	QTY.	R.F PAD ODxTHK	REMARK
MH2	MANWAY	600	–/20 THK.	150	WNRF	1	ø1050 x 25 THK	WITH BLIND FLANGE AND DAVIT
MH1	MANWAY	600	–/20 THK.	150	WNRF	1	ø1050 x 20 THK	
N13A/B	SPARE	50	160/8.74	150	WNRF	2	–	WITH BLIND FLANGE
N12	VAPOUR RETURN	50	160/8.74	150	WNRF	1	–	WITH BLIND FLANGE
N11A/B	TEMPERATURE GAUGE	50	160/8.74	150	WNRF	2	–	WITH BLIND FLANGE
N10	ROCHESTER GAUGE	50	160/8.74	150	WNRF	1	–	WITH BLIND FLANGE
N9	PRESSURE GAUGE	50	160/8.74	150	WNRF	1	–	WITH BLIND FLANGE
N8	FIXED LEVEL GAUGE	50	160/8.74	150	WNRF	1	–	WITH BLIND FLANGE
N7	VENT	50	160/8.74	150	WNRF	1	–	WITH BLIND FLANGE
N6A/B/C	SAFETY VALVE	50	160/8.74	150	WNRF	3	–	WITH BLIND FLANGE
N5	DRAIN	50	160/8.74	150	WNRF	1	–	WITH BLIND FLANGE
N4	LIQUID RETURN	50	160/8.74	150	WNRF	1	–	WITH BLIND FLANGE
N3	VAPOUR OUTLET	80	80/7.62	150	WNRF	1	ø200 x 20 THK	WITH BLIND FLANGE
N2	LIQUID OUTLET	80	80/7.62	150	WNRF	1	ø200 x 20 THK	WITH BLIND FLANGE
N1	LIQUID INLET	80	80/7.62	150	WNRF	1	ø200 x 20 THK	WITH BLIND FLANGE

NOZZLE DATA (FOR ONE EQUIPMENT ONLY)



NOZZLE NO.	SIZE (DN)	FORCE FA (N)	FORCE FL (N)	FORCE FC (N)	FORCE FR (N)	MOMENT MT (N-m)	MOMENT ML (N-m)	MOMENT MC (N-m)	MOMENT MR (N-m)
N1,N2,N3	80	±3600	3600	2700	4500	1098	883	657	1009.9
N4,N6A,N6B,N6C, N12,N13A,N13B	50	±2400	2400	1800	3000	490	392	294	490

NOZZLE LOADS TABLE

REFERENCED CODES AND STANDARDS

TITLE	CODE / STANDARD	APPLICABLE EDITION
SEAMLESS PIPE	ASME B36.10M	EDITION 2018
NON METALIC GASKETS FOR PIPE FLANGES	ASME B16.21	EDITION 2017
FLANGES	ASME B16.5	EDITION 2013
PR.PART MATERIALS (FERROUS)	ASME SEC.II PART A	EDITION 2019
WELDING CONSUMABLES	ASME SEC.II PART C	EDITION 2019
MATERIAL PROPERTIES	ASME SEC.II PART D(M)	EDITION 2019
NDE PROCEDURES	ASME SEC.V	EDITION 2019
NDE PERSONNEL QUALIFICATION	SNT-TC-1A	EDITION 2006
WELDING PROCEDURE AND PERFORMANCE QUALIFICATIONS.	ASME SEC.IX	EDITION 2017
SEISMIC CODE	NA	
WIND CODE	–	–

NON-DESTRICTIVE EXAMINATION TABLE NO.1

RADIOGRAPHY & JOINT EFFICIENCY: RT-1 & UG-116(a)(1)

WELD CATEGORY AS PER UW-3	COMPONENT	WELD JOINT TYPE NO. AS PER TABLE UW-12	RADIOGRAPHY EXTENT	CLAUSE	JOINT EFFICIENCY AS PER TABLE UW-12
CATEGORY A	SHELL	TYPE NO. 1	FULL	UW-11(a)	1.00
	HEAD	TYPE NO. 1	FULL	UW-11(a)	1.00
	FABRICATED NOZZLE	TYPE NO. 1	FULL	UW-11(a)	1.00
CATEGORY B	SHELL	TYPE NO. 1	FULL	UW-11(a)	1.00
	SHELL TO HEAD	TYPE NO. 1	FULL	UW-11(a)	1.00
CATEGORY C	FOR ALL NOZZLES	TYPE NO. 1	FULL FOR MH1 & MH2 (REMAINING ARE EXEMPTED)	UW-11(a)	1.00
CATEGORY D	FOR ALL NOZZLES	TYPE NO. 7	NA	–	–
VISUAL EXAMINATION	(INSIDE & OUTSIDE)				
CATEGORY A & B	SHELL & HEAD	TYPE NO. 1	FULL	UW-35	–
CATEGORY C	FOR ALL NOZZLES	TYPE NO. 1	FULL	UW-35	–
CATEGORY D	FOR ALL NOZZLES	TYPE NO. 7	FULL	UW-16	–
PENETRANT EXAMINATION	1. 100% DP TESTING OF INSIDE AND OUTSIDE SURFACE OF KNUCKLE AREA AND SF AREA ALL LOAD BEARING FILLET WELDS.				
MAGNETIC EXAMINATION	NOT APPLICABLE				
ULTRASONIC EXAMINATION	NOT APPLICABLE				

MATERIAL SPECIFICATION

SHELL/HEAD		SA-516 GR.70 N	MANHOLE	NECK (PLATE)	SA-516 GR.70 N
REINFORCEMENT PADS		SA-516 GR.70 N		FLANGE/COVER	SA-105 N
EXTERNAL BRANCHES	NECK (PIPE/PLATE)	SA-106 GR. B(N)/SA-516 GR.70 N		STUDS/NUTS	SA-193 GR. B7 / SA-194 GR. 2H
	FLANGE/BLIND FLANGE	SA-105 N		GASKET	NACF (m=2.0 , y=11.03 MPa)
	STUDS/NUTS	SA-193 GR. B7/SA-194 GR. 2H		HANDLE	SA-516 GR.70 N
	GASKET	NACF (m=2.0 , y=11.03 MPa)	DAVIT	PIPE	SA-106 GR. B
SADDLE SUPPORT	WEAR PLATE	SA-516 GR.70 N		CLEAT/SUPPORT	SA-516 GR.70 N
	RIB PLATE/WEB PLATE	IS 2062 GR. A		EYE BOLT/NUTS	SA-193 GR. B7/SA-194 GR. 2H
	BASE PLATE	IS 2062 GR. A	NAME PLATE	NAME PLATE	SS-304
LIFTING LUG (PAD/PLATE)		SA-516 GR.70 N		BRACKET	SA-516 GR.70 N
ANCHOR BOLT (BY OTHERS)		IS 2062 (HDG) ALLOWABLE TENSILE STRENGTH -114 MPa, SHEAR STRESS -100 MPa.		RIVETS	SS-304
				EARTHING LUG	SS-304

FOUNDATION LOADING DATA

LOAD CONDITION	SLIDING SADDLE VERTICAL LOAD (KN)	SLIDING SADDLE SHEAR FORCE (KN) LONG.	SLIDING SADDLE SHEAR FORCE (KN) TRANS.	SLIDING SADDLE BASE MOMENT (KN-m) LONG.	SLIDING SADDLE BASE MOMENT (KN-m) TRANS.	FIXED SADDLE VERTICAL LOAD (KN)	FIXED SADDLE SHEAR FORCE (KN) LONG.	FIXED SADDLE SHEAR FORCE (KN) TRANS.	FIXED SADDLE BASE MOMENT (KN-m) LONG.	FIXED SADDLE BASE MOMENT (KN-m) TRANS.
OPERATING (WIND)	725	11.32	9.450	20.1	16.77	725	11.32	9.450	20.1	16.77
FRICTION LOAD		117								

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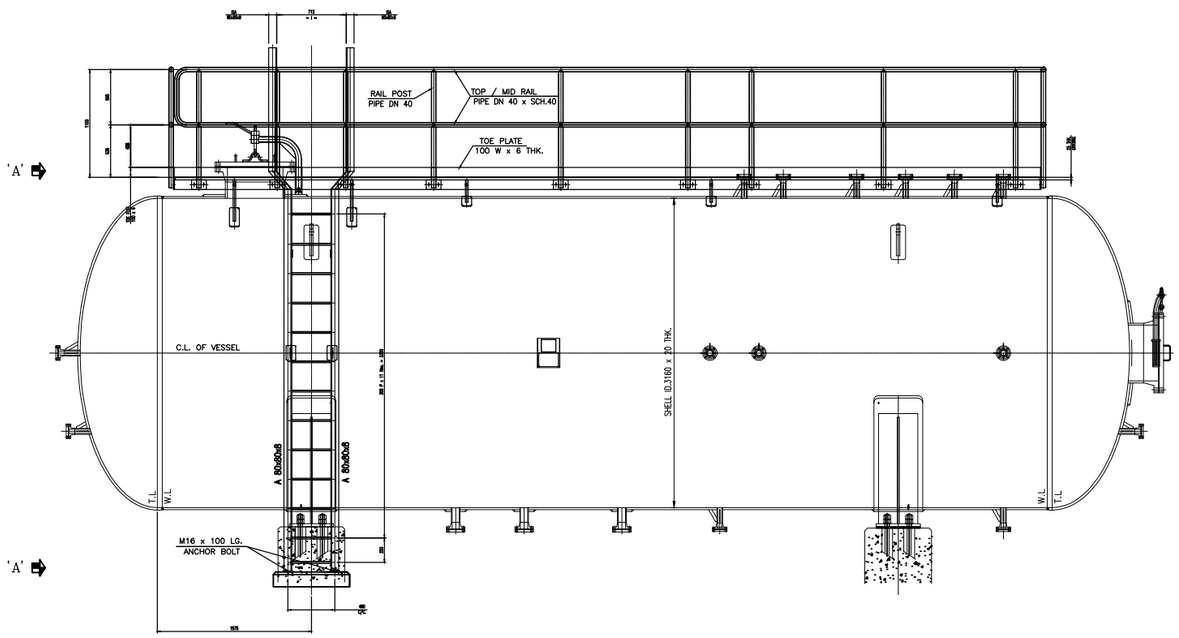
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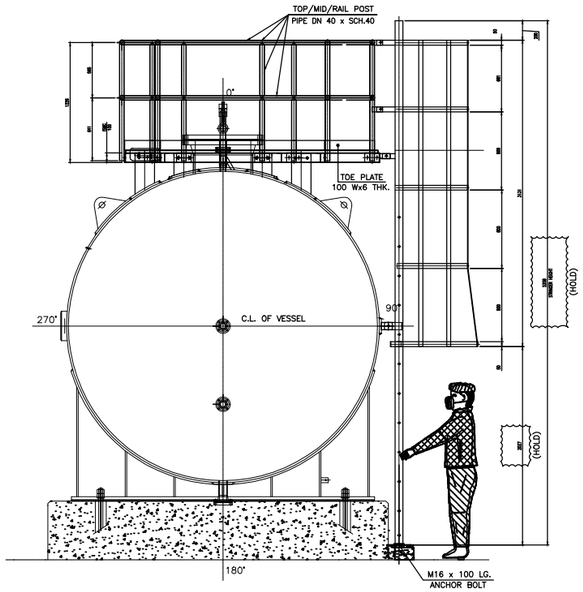
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DATE	26.09.2020	DATE	26.09.2020	DATE	26.09.2020
DRAWN	GB	CHECKED	NM	APPROVED	SAR

L&T OUTOTEC DRAWING NUMBER	019045_080_U_V_19163_001_R1		REVISION 01
OUTOTEC DRAWING NUMBER	900707__GAU01__AG50__MTL01__49003	DOCUMENT ID OU100688548	REVISION 00

DRAWN	---	26.09.2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	MNP	26.09.2020	GENERAL ARRANGEMENT FUEL OIL STORAGE
ENGINEER	---	26.09.2020	FUEL OIL STORAGE TANK
APPROVED	QRS	26.09.2020	LPG BULLET
EMPLOYER			
SCALE	1:XX	A0	DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1003 REV: A01



ELEVATION



VIEW FROM 'A'–'A'

NOTES :–

1) ALL DIMENSIONS ARE IN mm. UNLESS OTHERWISE SPECIFIED.
2) PLATFROM & LADDER CONSTRUCTION IS AS PER GOOD ENGINEERING PRACTISE. AND AS PER LOADINGS.
3) PLATFORM ELEVATIONS, SHOWN ON DRAWINGS, ARE TO TOP OF GRATING.
4) REMOVABLE GRATING PANEL SHALL BE FIXED WITH SUPPORTING MEMBERS BY CLAMP ONLY.
5) ALL PADS HAVE 1 No. Ø6 VENT HOLE.
6) LENGTH OF MEMBER AND SIZE OF CLEAT SHALL BE VERIFIED AT SHOP BY FULL SCALE LAYOUT PRIOR TO FABRICATION.
7) STEEL STRUCTURE PLATFORM SHALL BE SHOP ASSEMBLED ON THE EQUIPMENT AT WORK. SHOP TO VERIFY CORRECT FITTING.
8) BEFORE SURFACE PREPARATION SHALL REMOVE ALL SHARP CORNERS,BURRS IN BOLT HOLES. WELD SPATTER SLAGS,WELD FLUX , SCALE AND OTHER FOREIGN MATTERS.
9) ALL CUT CORNERS SHALL BE GROUND SMOOTH.
10) ALL PLATFORM COMPONENTS (SUPPORT BEAMS, HANDRAILS, GRATINGS AND LADDER) SHALL BE PIECE MARKED AND/MATCH MARKED AFTER PAINTING AND SHALL BE REMOVED FROM VESSEL SEPARATELY FOR SHIPMENT TO SITE.
11) PAINTING SHALL BE PERFORMED AFTER TRIAL FIT–UP ASSEMBLY
12) EACH TAG MARK SHALL BE PREFIXED WITH EQUIPMENT MARK.
13) SURFACE FINISH FOR GRATING SURFACE SHALL BE CLEAN & FREE RUST OIL, DIRT, SCALE, OXIDES & FORGEN MATERIAL DETERMENTAL
14) ATTACHMENT WELDED TO THE EQUIPMENT SHALL NOT BE NOTCHED TO CLEAR CIRCUMFERENTIAL WELD SEAMS ON THE EQUIPMENT WITHOUT APPROVAL BY THE ENGINEER OF RECORD.

PAINTING SYSTEM:

STRUCTURAL STEEL FRAME, KICK PLATE, LADDER, CAGE, PLATFORM, GUARD RAILS, HANDRAILS

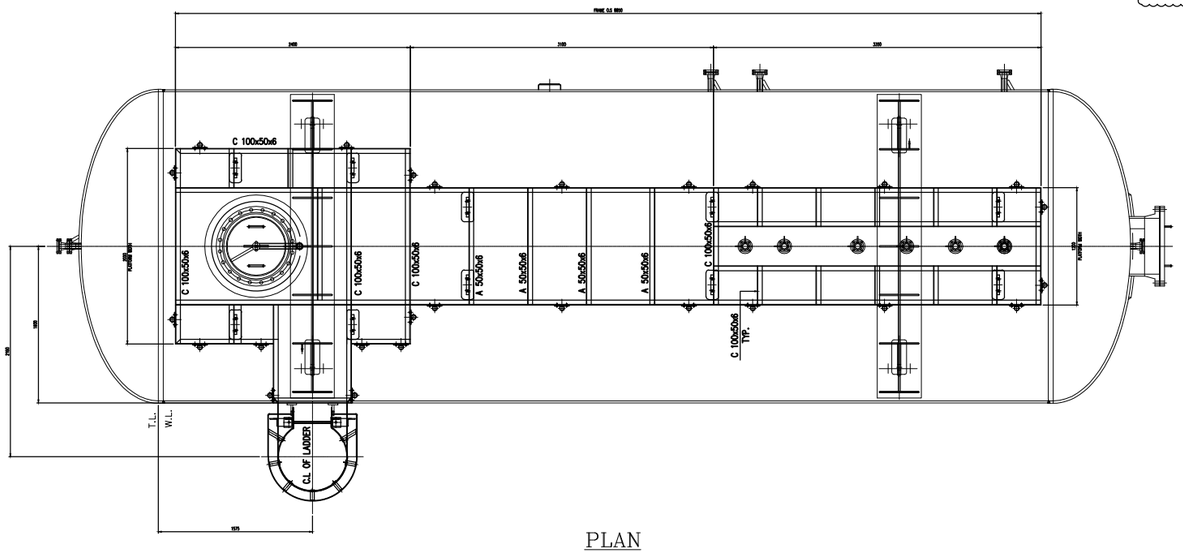
SURFACE PREPARATION	: SA 2.5
PRIMER	: ZINC EPOXY 1 COAT (80 DFT)
FINISH	: EPOXY HIGH SOLID 1 COAT (80 DFT)
TOTAL DFT	: 2 COATS (160 DFT)
FINISHING COLOUR	: SIGNAL YELLOW RAL 1003
GRATINGS	: HOT DIP GALVANIZED (STD. DFT)

LEGEND: –

T.O.G. – TOP OF GRATING
– MILD STEEL GRATING
– RECTANGULAR / SQUARE OPENING CUTOUT
– CIRCULAR OPENING CUTOUT
– DIRECTION OF SPAN OF GRATING/ ONE WAY SLAB

T.O.S. – TOP OF STEEL
T.O.G. – TOP OF GRATING
PL. – PLATFORM
EL. – ELEVATION
T.L. – TAN LINE
W.L. – WELD LINE

MARKED DIMENSIONS ARE HOLD TILL CONFIRMATION OF SITE PLAN, CIVIL FOUNDATION / BOLTS DETAILS & DIMENSIONS.
(HOLD)



PLAN

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	---	MNP	---	QRS	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER						REVISION
2006-UNI-DWG-076						01
DATE	26.09.2020	DATE	26.09.2020	DATE	26.09.2020	
DRAWN	GB	CHECKED	NP	APPROVED	SKR	

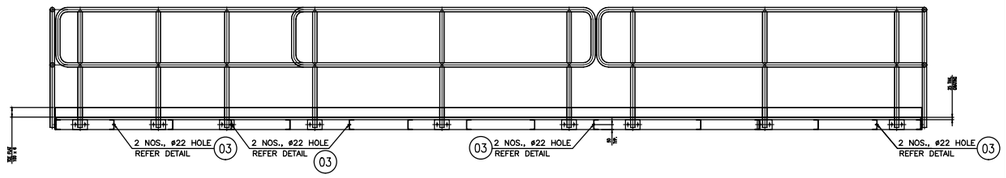
LARSEN & TOUBRO DRAWING NUMBER: O1904 5_080_U_V_19163_001_R1 — REVISION 01
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG50__MTL01__49003 — DOCUMENT ID: OU100688548 — REVISION 00

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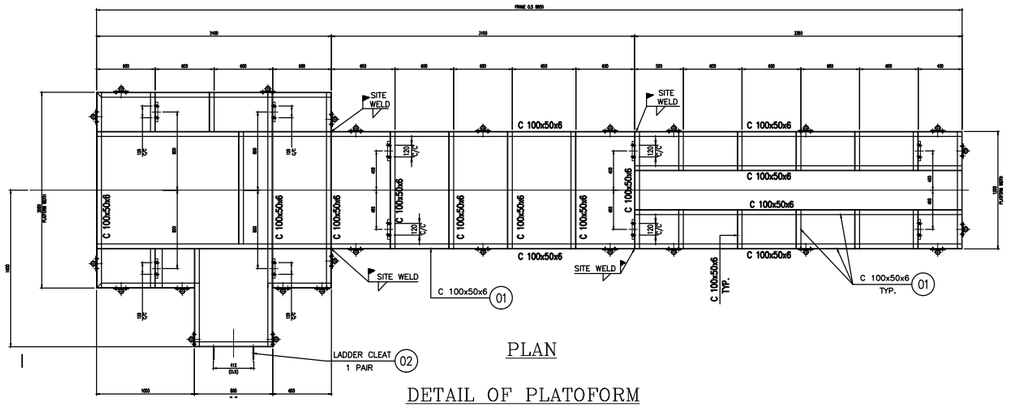
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DRAWN	---	26.09.2020	MANSOURAH – MASSARAH GOLD PROJECT
CHECKED	MNP	26.09.2020	GENERAL ARRANGEMENT FUEL OIL STORAGE
ENGINEER	---	26.09.2020	FUEL OIL STORAGE TANK
APPROVED	QRS	26.09.2020	LPG BULLET
EMPLOYER			
SCALE 1:XX		A0	DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1003 — REV: A01

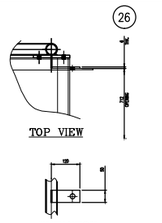


ELEVATION

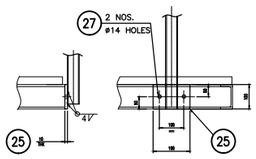


PLAN

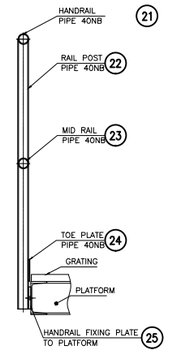
DETAIL OF PLATOFORM



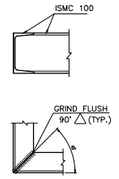
TOP HOOP FIXING CLEAT ON HANDRAIL



DETAIL OF 'B1' HANDRAIL FIXING PLATE



RAIL POST DETAIL ON PLATFORM



DETAIL CHANNEL TO CHANNEL FRAME CORNER JOINT

BILL OF MATERIAL–PLATFORM (FOR ONE EQUIPMENT ONLY)

PART No.	DESCRIPTION	QTY.	MATERIAL	SIZE	REMARK
01	CHANNEL FOR PLT_BKT–1	02	I.S 2062	ISMC 100 x 50 x 6 x ~42000	–
02	LADDER SUPPORT CLEAT	02	SA–516 GR.70	130 x 70 x 6 THK	–
03	BOLTS, NUTS WITH 2 WASHER	20	SA–193 GR. B7 / SA–194 GR.7	M20 x 55 LG.	HOT DIP GALVANIZED

BILL OF MATERIAL–HANDRAIL(FOR ONE EQUIPMENT ONLY)

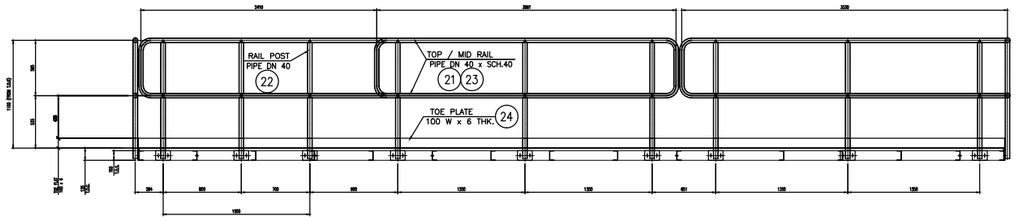
PART No.	DESCRIPTION	QTY.	MATERIAL	SIZE	REMARK
21	HAND RAIL	01	SA–106 GR.B	PIPE 40NB x SCH.40 x ~42000 LG.	–
22	HANDRAIL POST	01	SA–106 GR.B	PIPE 40NB x SCH.40 x ~32500 LG.	–
23	MID RAIL	01	SA–106 GR.B	PIPE 40NB x SCH.40 x ~42000 LG.	–
24	TOE PLATE	01	SA–516 GR.70	PLATE 100 x 6 THK x ~42000 LG.	–
25	RAIL POST FIXING PLATE	26	SA–516 GR.70	PLATE 150 x 100 x 10THK.	–
26	CLEAT FOR LADDER HOOP FIXING	04	SA–516 GR.70	~120 x 50 x 6 THK	–
27	BOLTS, NUTS WITH 2 WASHER	52	SA–193 GR. B7 / SA–194 GR.7 (HDG.)	M12 x 55 LG.	HOT DIP GALVANIZED

BILL OF MATERIAL–GRATING (FOR ONE EQUIPMENT ONLY)

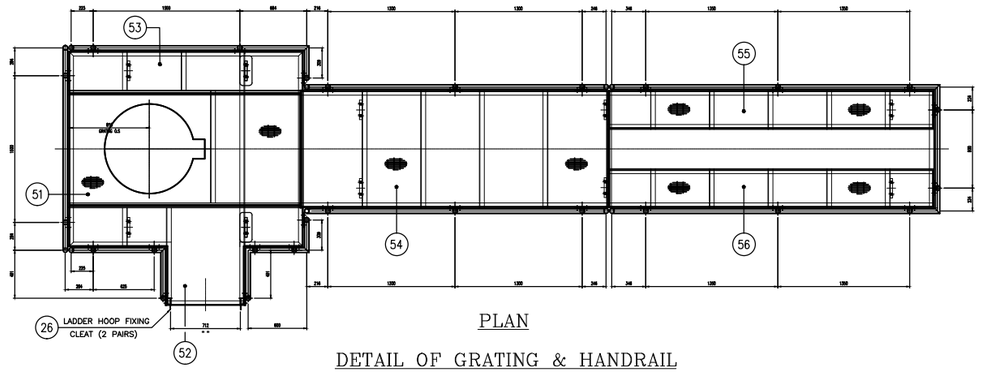
PART No.	DESCRIPTION	QTY.	MATERIAL	SIZE	REMARK
51	GRATING–1	01	CS (GALV.)	1135 x 2360 x 25 THK	–
52	GRATING–2	01	CS (GALV.)	2300 x 1010 x 25 THK	AS PER SHAPE
53	GRATING–3	01	CS (GALV.)	2380 x 415 x 25 THK	–
54	GRATING–4	01	CS (GALV.)	3140 x 1180 x 25 THK	–
55	GRATING–5	01	CS (GALV.)	3305 x 375 x 25 THK	–
56	GRATING–6	01	CS (GALV.)	3305 x 375 x 25 THK	–
57	CLAMP SET FOR GRATING	25	CS (GALV.)	STD. TO SUIT, AS PER DETAIL	–

NOTES :–

1) ALL DIMENSIONS ARE IN mm. UNLESS OTHERWISE SPECIFIED.
2) PLATFORM, LADDER & HOOP FIXING CLEATS SHALL BE WELDED IN ASSEMBLY CONDITION TO MATCH ON SITE DURING ERECTION.

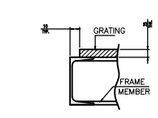


ELEVATION

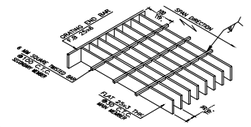


PLAN

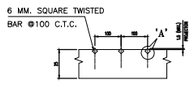
DETAIL OF GRATING & HANDRAIL



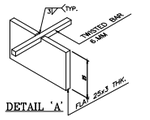
DETAIL OF GRATING SPACING FOR PLATFORM OUTER EDGE



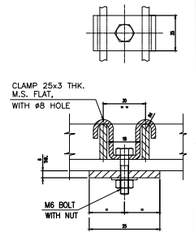
GRATING DETAIL



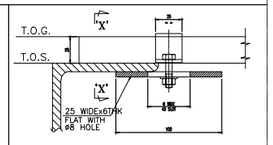
GRATING DETAIL



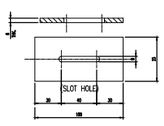
DETAIL 'A'



57 VIEW 'X–X' CLAMP DETAIL AT CENTER (AT WITHOUT SUPPORTING SIDE)



CLAMP DETAIL AT CORNER



DETAIL OF CLAMP FIXED PLATE

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-ERT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	---	MNP	---	QRS	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER		REVISION
2006-UNI-DWG-076		01
DATE 26.09.2020	DRAWN GB	CHECKED NM	APPROVED SKR

LARSEN & TOUBRO DRAWING NUMBER: O19045_080_U_V_19163_001_R1 — REVISION 01
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG50__MTL01__49003 — DOCUMENT ID: OU100688548 — REVISION 00

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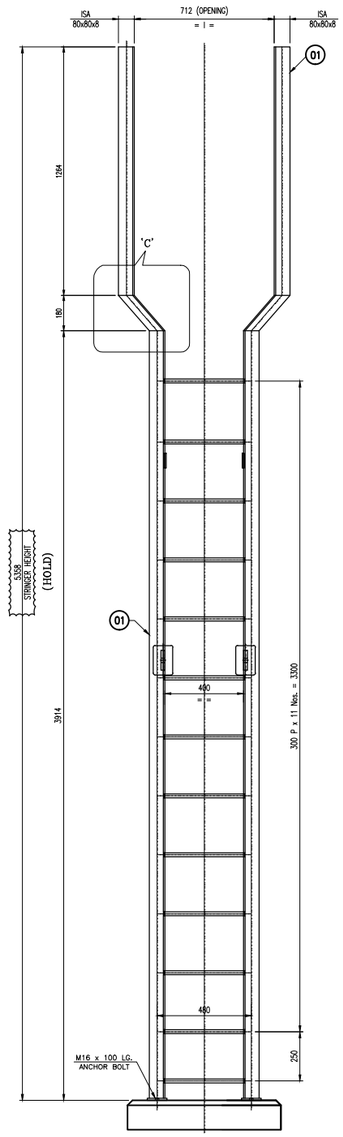
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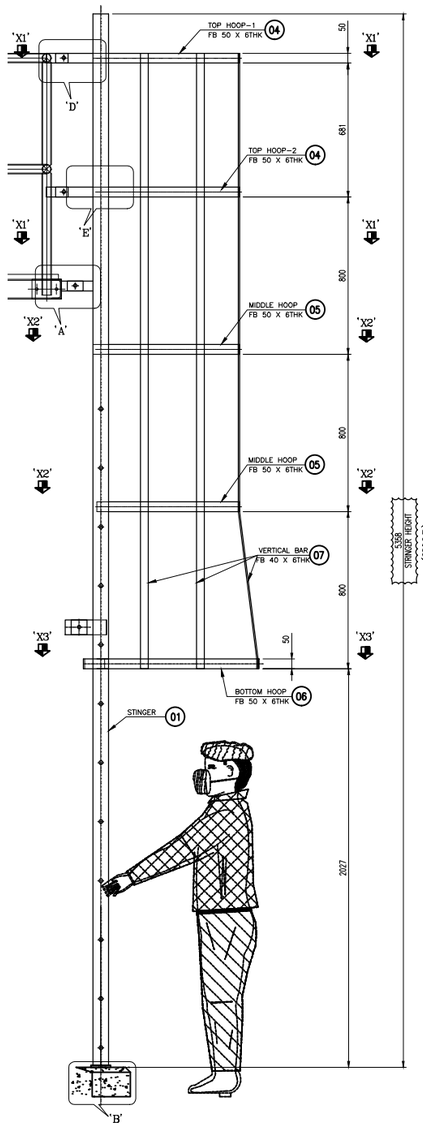
DRAWN	---	26.09.2020	MANSOURAH – MASSARAH GOLD PROJECT
CHECKED	MNP	26.09.2020	GENERAL ARRANGEMENT FUEL OIL STORAGE
ENGINEER	---	26.09.2020	FUEL OIL STORAGE TANK
APPROVED	QRS	26.09.2020	LPG BULLET
EMPLOYER			

SCALE 1:XX — A0

DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1003 — REV A01

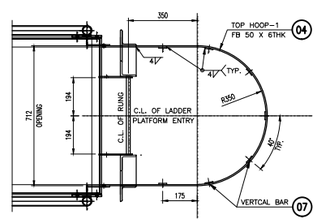


FRONT VIEW OF LADDER WITHOUT CAGE
(FOR CLARITY)

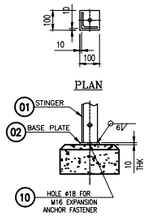


SIDE VIEW

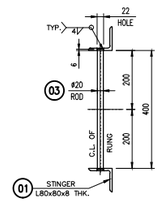
DETAIL OF LADDER



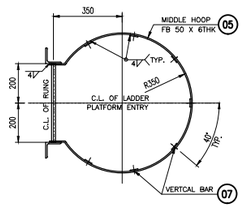
TOP HOOP
VIEW FROM 'X1-X1'



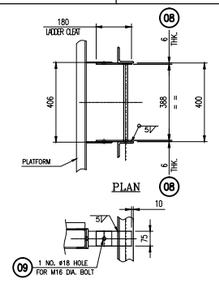
DETAIL-'B'



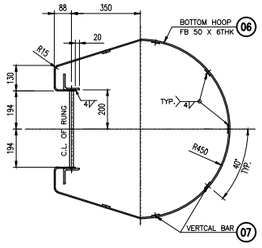
DETAIL OF LADDER RUNG



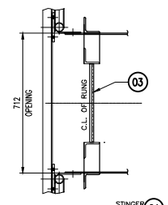
MIDDLE HOOP
VIEW FROM 'X2-X2'



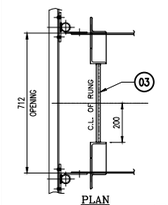
DETAIL-'A'
FIXED TYPE CLEAT (F.T.)



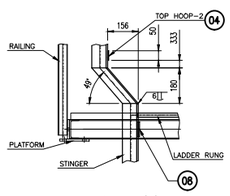
BOTTOM HOOP
VIEW FROM 'X3-X3'



DETAIL-'D'



DETAIL-'E'



DETAIL-'C'

BILL OF MATERIAL
(FOR ONE EQUIPMENT ONLY)

PART No.	DESCRIPTION	QTY.	MATERIAL	SIZE	REMARK
01	LADDER STINGER	02	I.S 2062	L80x80x8 THK.x5450 LG.(CUT TO SUIT)	–
02	LADDER BASE PLATE	02	SA-516 GR.70	100 x 100 x 10 THK.	–
03	BAR RUNG	12	I.S 2062	#20 BAR X 408 LG.	–
04	TOP HOOP-1 & 2	02	I.S 2062	FB 50 x 6 THK x 2150	–
05	MIDDLE HOOP	02	I.S 2062	FB 50 x 6 THK x 1955	–
06	BOTTOM HOOP	01	I.S 2062	FB 50 x 6 THK x 2768 LG.	–
07	VERTICAL BAR	07	I.S 2062	FB 40 x 6 THK x 2460 LG.(CUT TO SUIT)	–
08	LADDER CLEAT-1	02	SA-516 GR.70	180 x 75 x 6 THK.	–
09	BOLTS, NUTS AND 2 WASHER	08	SA-193 GR. B7 / SA-194 GR.7	M16 x 55 LG.	HOT DIP GALVANIZED
10	ANCHOR BOLT	02	SA-193 GR. B7	M16 x 100 LG.	HOT DIP GALVANIZED
11	SAFETY CHAIN	02	I.S 2062	#6 x AS REQUIRED.	–

NOTES :-

1) ALL DIMENSIONS ARE IN mm. UNLESS OTHERWISE SPECIFIED.
2) PLATFORM, LADDER & HOOP FIXING CLEATS SHALL BE WELED IN ASSEMBLY CONDITION TO MATCH ON SITE DURING ERECTION.

MARKED DIMENSIONS ARE HOLD TILL CONFIRMATION OF SITE CIVIL FOUNDATION / BOLTS DETAILS & DIMENSIONS.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-ERT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A0X	XX/XX/XXXX	ISSUED FOR APPROVAL / REFERENCE / INFORMATION	---	MNP	---	QRS	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER: 2006-UNI-DWG-076 — REVISION 01
DATE 26.09.2020 | DRAWN GB | DATE 26.09.2020 | CHECKED NM | DATE 26.09.2020 | APPROVED SKR

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_V_19163_001_R1 — REVISION 01
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ AG50_ MTL01 _ 49003 — DOCUMENT ID: OU100688548 — REVISION 00

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DRAWN	---	26.09.2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	MNP	26.09.2020	GENERAL ARRANGEMENT FUEL OIL STORAGE
ENGINEER	---	26.09.2020	FUEL OIL STORAGE TANK
APPROVED	QRS	26.09.2020	LPG BULLET
EMPLOYER			DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1003 — REV A01
SCALE	1:XX	A0	

ATTACHMENT 2 THERMAL EXPANSION COEFFICIENT

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Total linear expansion between 70°F and indicated temperature (inches/100 ft)

Temp. (°F)	Shell Material								Temp. (°F)
	Carbon Steel Carbon - Moly Low-Chrome (through 3 Cr Mo)	5 Cr Mo through 9 Cr Mo	Austenitic Stainless Steels 18 Cr 8 Ni	12 Cr 17 Cr 27 Cr	12 Cr 20 Ni	Monel 67 Ni 30 Cu	3-1/2 Nickel	Ni-Fe-Cr	
70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70
100	0.23	0.22	0.34	0.20	0.32	0.28	0.23	0.28	100
125	0.42	0.40	0.62	0.36	0.58	0.52	0.42	0.52	125
150	0.61	0.58	0.90	0.53	0.84	0.75	0.61	0.76	150
175	0.80	0.76	1.18	0.69	1.10	0.99	0.81	0.99	175
200	0.99	0.94	1.46	0.86	1.37	1.22	1.01	1.23	200
225	1.21	1.13	1.75	1.03	1.64	1.46	1.21	1.49	225
250	1.40	1.33	2.03	1.21	1.91	1.71	1.42	1.76	250
275	1.61	1.52	2.32	1.38	2.18	1.96	1.63	2.03	275
300	1.82	1.71	2.61	1.56	2.45	2.21	1.84	2.30	300
325	2.04	1.90	2.90	1.74	2.72	2.44	2.05	2.59	325
350	2.26	2.10	3.20	1.93	2.99	2.68	2.26	2.88	350
375	2.48	2.30	3.50	2.11	3.26	2.91	2.47	3.18	375
400	2.70	2.50	3.80	2.30	3.53	3.25	2.69	3.48	400
425	2.93	2.72	4.10	2.50	3.80	3.52	2.91	3.76	425
450	3.16	2.93	4.41	2.69	4.07	3.79	3.13	4.04	450
475	3.39	3.14	4.71	2.89	4.34	4.06	3.35	4.31	475
500	3.62	3.35	5.01	3.08	4.61	4.33	3.58	4.59	500
525	3.86	3.58	5.31	3.28	4.88	4.61	3.81	4.87	525
550	4.11	3.80	5.62	3.49	5.15	4.90	4.04	5.16	550
575	4.35	4.02	5.93	3.69	5.42	5.18	4.27	5.44	575
600	4.60	4.24	6.24	3.90	5.69	5.46	4.50	5.72	600
625	4.86	4.47	6.55	4.10	5.96	5.75	4.74	6.01	625
650	5.11	4.69	6.87	4.31	6.23	6.05	4.98	6.30	650
675	5.37	4.92	7.18	4.52	6.50	6.34	5.22	6.58	675
700	5.63	5.14	7.50	4.73	6.77	6.64	5.46	6.88	700
725	5.90	5.38	7.82	4.94	7.04	6.94	5.70	7.17	725
750	6.16	5.62	8.15	5.16	7.31	7.25	5.94	7.47	750
775	6.43	5.86	8.47	5.38	7.58	7.55	6.18	7.76	775
800	6.70	6.10	8.80	5.60	7.85	7.85	6.43	8.06	800
825	6.97	6.34	9.13	5.82	8.15	8.16	6.68	8.35	825
850	7.25	6.59	9.46	6.05	8.45	8.48	6.93	8.66	850
875	7.53	6.83	9.79	6.27	8.75	8.80	7.18	8.95	875
900	7.81	7.07	10.12	6.49	9.05	9.12	7.43	9.26	900
925	8.08	7.31	10.46	6.71	9.35	9.44	7.68	9.56	925
950	8.35	7.56	10.80	6.94	9.65	9.77	7.93	9.87	950
975	8.62	7.81	11.14	7.17	9.95	10.09	8.17	10.18	975

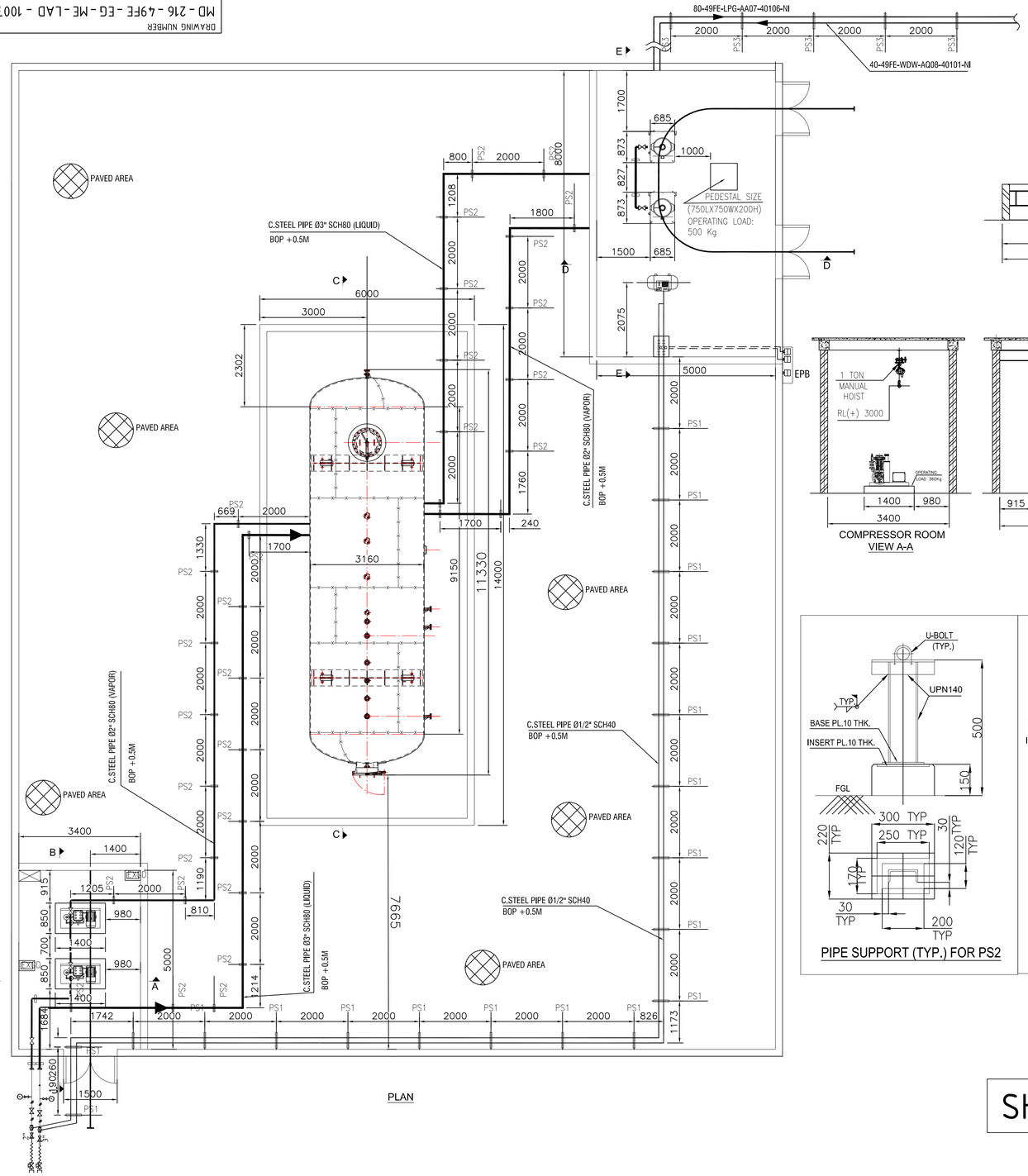
ATTACHMENT 3 MISCELLANEOUS PIPE SUPPORT DESIGN

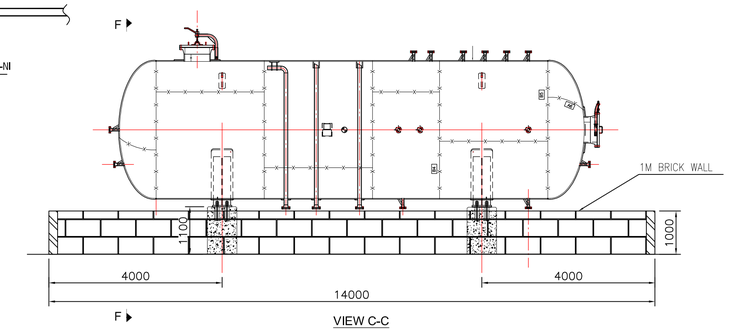
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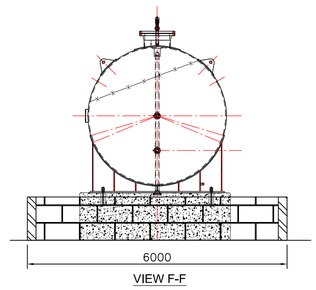
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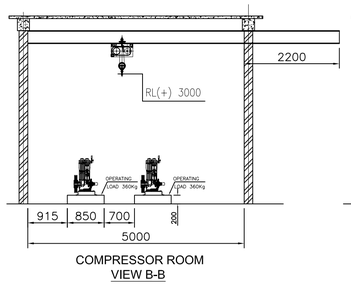
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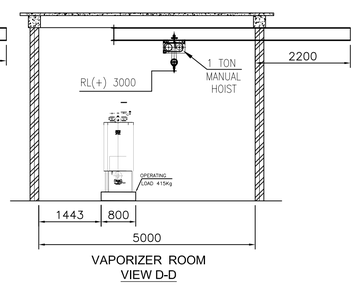
DRAWING NUMBER MD-216-49FE-EG-ME-LAD-1007 REV A01

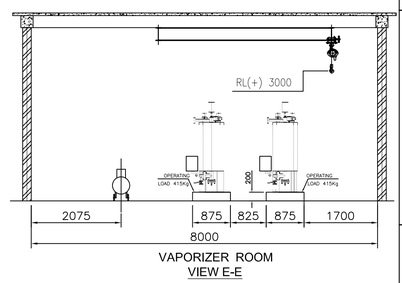


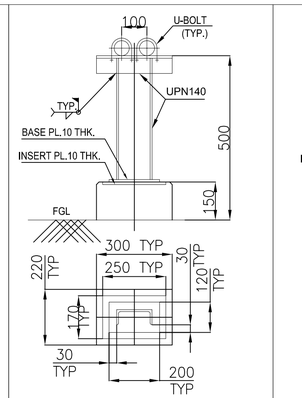


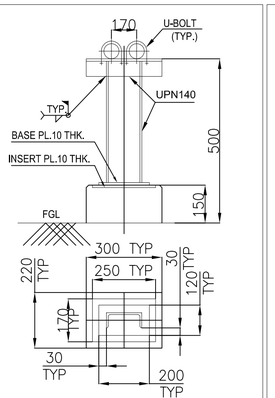












PIPE LOAD WITH WATER

S.NO.	PIPE DIA NB	PIPE WEIGHT WITH WATER KG/M
1.	15	1.2
2.	20	2
3.	25	3
4.	40	5.3
5.	50	8
6.	65	12
7.	80	16

NOTE :
1. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M UNLESS OTHERWISE SPECIFIED.
2. 0.00 LEVEL CORRESPONDS TO FINISHED GROUND LEVEL.
3. DIMENSIONS AND SUPPORTS LOCATION WHICH EVER IS REQUIRED IF NECESSARY BESIDES THE SUPPORTS INDICATED IN DRAWING MAY BE DECIDED SUITABLY AT SITE.
4. PIPE OPENING TO BE PROVIDED WITH PUDDLE FLANGE.
5. REFER DRAWING: LPG UNLOADING, STORAGE & TRANSFER SYSTEM.
6. REFER DRAWING: LPG UNLOADING, STORAGE & TRANSFER SYSTEM - COMPRESSOR AND MOTOR SKID GA DRAWINGS, VAPORIZER GA DRAWINGS LPG BULLET GA DRAWING.

SHEET-2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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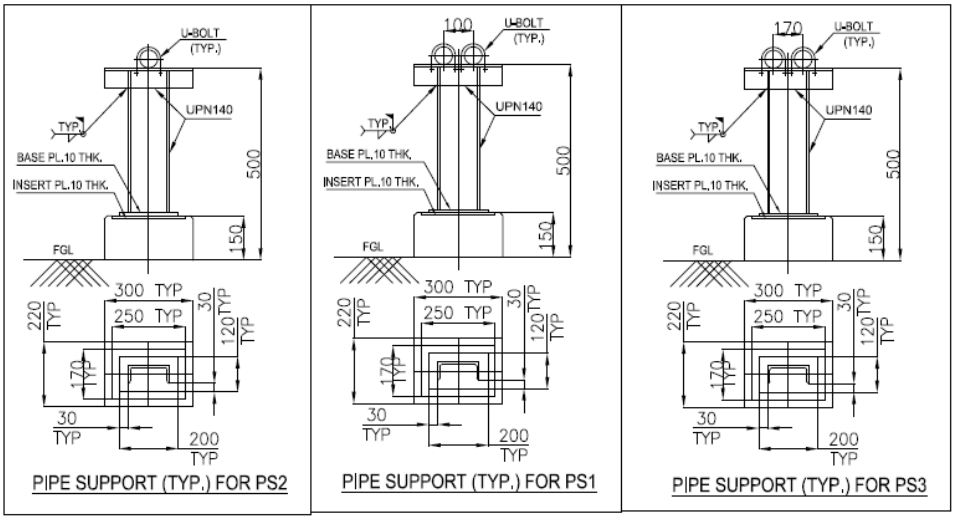
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MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
FUEL OIL STORAGE
EQUIPMENT ARGT. WITH PIPING FOR LPG SYSTEM

DRAWING NUMBER MD-216-49FE-EG-ME-LAD-1007 REV A01

PIPE SUPPORT TYPE:



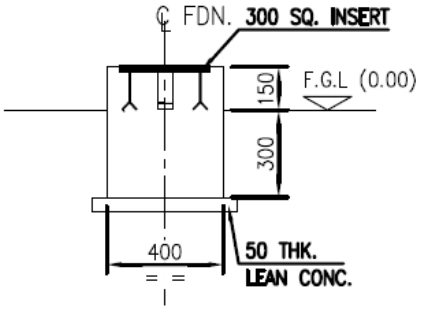
PIPE SUPPORT LOADINGS:

PIPE LOAD WITH WATER

S.NO.	PIPE DIA NB	PIPE WEIGHT WITH WATER KG/M
1.	15	1.2
2.	20	2
3.	25	3
4.	40	5.3
5.	50	8
6.	65	12
7.	80	16

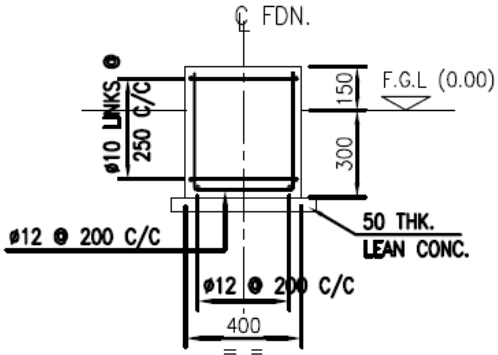
As the forces are minimal and the pipes are located close to ground surface, no separate calculation performed for these supports. Support details are provided based on engineering judgment.

Refer next page for support detail.



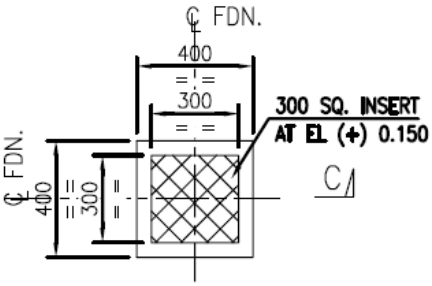
G.A. SECTION C-C

SCALE 1:20



R.C. SECTION C-C

SCALE 1:20



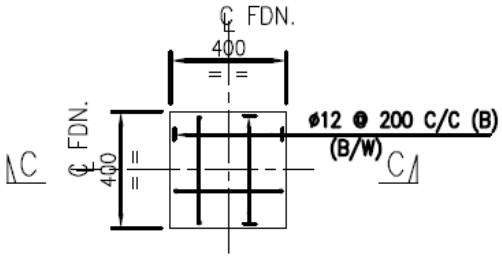
PLAN

DETAIL OF

FOUNDATION MKD. F2

(50-NOS. REQUIRED)

SCALE 1:20



PLAN

R.C. DETAIL OF

FOUNDATION MKD. F2

(50-NOS. REQUIRED)

SCALE 1:20

Refer drawing number: MD-216-49FE-EG-CV-GAD-1631 for support location.